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On the threshold dividend strategy for a generalized jump–diffusion risk model

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ABSTRACT

In this paper, we generalize the Cramér–Lundberg risk model perturbed by diffusion to incorporate jumps due to surplus fluctuation and to relax the positive loading condition. Assuming that the surplus process has exponential upward and arbitrary downward jumps, we analyze the expected discounted penalty (EDP) function of Gerber and Shiu (1998) under the threshold dividend strategy. An integral equation for the EDP function is derived using the Wiener–Hopf factorization. As a result, an explicit analytical expression is obtained for the EDP function by solving the integral equation. Finally, phase-type downward jumps are considered and a matrix representation of the EDP function is presented.

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1. Introduction

The ruin probabilities and more generally the expected discounted penalty (EDP) function for the Cramér–Lundberg risk model without and with diffusion have been studied extensively—see e.g. Dufresne and Gerber (1991), Gerber and Shiu (1998), Gerber and Landry (1998), Lin and Willmot (1999) and Tsai and Willmot (2002), and references therein. Many analytical results on the ruin probabilities and the EDP function under these models are obtainable due to the property of stationary and independent increments of the Poisson process and Brownian motion. These models and the analysis of the ruin probabilities and the EDP function have since been extended in several ways. One natural extension is to consider a spectrally negative Lévy process for the insurer's surplus, in which only downward jumps are allowed as in Yang and Zhang (2001) and Garrido and Morales (2006).

When a risk model has a diffusion component, it is used to model the fluctuation of the surplus due to the investment of the surplus in the equity market. However, its normality behavior is insufficient to explain the leptokurtic feature in equity prices, which is reconfirmed by a recent empirical study in Aït-Sahalia and Jacod (2009) that provides evidence for the presence of jumps in equity indices. A double exponential jump–diffusion equity model was introduced in Kou (2002) to incorporate jumps into a diffusion model, in order to partly overcome the shortcomings of a diffusion model and in the meantime to maintain the tractability of the model. Following the idea of Kou (2002), Chi (2010) assumes that the evolution of the surplus follows a double exponential jump–diffusion process. More recently, the analysis of the EDP function for a jump–diffusion risk model with two-sided arbitrary jumps and without dividend strategies is studied in Zhang et al. (2010) and Albrecher et al. (in press). The former uses an integro-differential equation approach and the latter offers a direct approach. Both still assume the smoothness of the EDP function implicitly.

Dividend strategies for an insurance risk model were first proposed by de Finetti (1957) to more realistically model the insurer's surplus. In the past decade, there has been enormous amount of interest in studying the EDP function for a variety of insurance risk models under the barrier and threshold dividend strategies. For more details, see Lin et al. (2003), Zhou (2005), Frostig (2005), Lin and Pavlova (2006), Gerber and Shiu (2006), Wan (2007), Frostig (2008), and references therein. Interestingly, Zhou (2005) makes use of the scale function associated with a Lévy process to obtain a representation of the EDP function for the Cramér–Lundberg risk model with the barrier dividend strategy. To the best of our knowledge, Zhou (2005) is the first paper to point out the connections between the fluctuation theory of Lévy processes and the EDP function of a risk model under a dividend strategy.

The study of the fluctuation theory of Lévy processes has a long history and results in many useful results. See Bertoin (1996), Sato (1999) and Kyprianou (2006), for example. Furthermore, Bingham (1975) studies one-sided and two-sided exit problems for a general Lévy process and obtains certain Wiener–Hopf factorizations and related factorization identities. A two-sided exit problem is much more complicated than its one-sided counterpart.

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For a two-sided exit problem, Kadankov and Kadankova (2005) obtain an expression of the joint distribution of the exit time from an interval and the corresponding overshoot. However, only a few papers have been dedicated to Lévy processes with both upward and downward jumps. Kou and Wang (2003) consider a double exponential jump–diffusion process and obtain some simple and interesting results for the one-sided exit problem. Kadankova and Veraverbeke (2007) consider a compound Poisson process with exponential upward jumps. They simplify the main result in Kadankov and Kadankova (2005) to obtain a closed-form expression for the two-sided exit problem. The methods developed for those exit problems have proved to be powerful in deriving analytical expressions for the EDP function of Lévy-type risk models under a dividend strategy.

In this paper, we study the EDP function for a risk model in which the surplus follows a Lévy process whose jump component is a compound Poisson process with exponential upward and arbitrary downward jumps. This is a generalization of the model in Kou (2002) and the classical compound Poisson model perturbed by diffusion originally proposed in Dufresne and Gerber (1991) but slightly less general than that in Zhang et al. (2010) and Albrecher et al. (in press). However, the threshold dividend strategy is incorporated in this paper such that dividends will be paid out at a constant rate as long as the insurer's retained surplus after deducting dividends is above a predetermined level (threshold). The applications of the method of Wiener–Hopf factorization to risk models can be dated back to Cramér (1954) and Täcklind (1942) as mentioned in Prabhu (1998) and Bühlmann (1970). The method has become very popular in the analysis of Lévy type risk models and can be found in many recent papers in ruin theory. See Asmussen (2000), Avram et al. (2003), Klüppelberg et al. (2004) and Zhou (2007), and references therein. Using the Wiener–Hopf factorization and an approach similar to that in Kadankova and Veraverbeke (2007), we derive an analytical expression of the joint distribution of the barrier hitting time and the overflow/deficit of the surplus process under consideration. We then use the results to derive an explicit analytical expression of the EDP function for the risk model without the assumption of the smoothness of the EDP function. When the downward jumps are phase-type distributed, our result is further simplified and the EDP function is of a nice matrix form.

This paper is organized as follows. Section 2 specifies the risk model mentioned above. Section 3 derives explicit expressions of the distributions of the supremum and infimum of the surplus process with no initial surplus that is killed at an independent exponential random time by the Wiener–Hopf factorization. In Section 4, we analyze the barrier hitting problems of the surplus process with the help of the results in Section 3. We obtain closed-form expressions for the joint distributions of the barrier hitting time and the corresponding overflow/deficit for both one barrier and two barrier cases. These results are critical in the analysis of the EDP function. Section 5 provides an explicit analytical expression of the EDP function, using the results in Section 4. In Section 6, phase-type downward jumps are considered and a matrix representation of the EDP function is given. Finally, some concluding remarks are made in Section 7.

2. Model specification

We consider an insurance risk model in which the insurer's surplus process is defined as

$$\tilde{U}_t = u + ct - \sum_{k=1}^{N_t} J_k + \sigma W_t, \quad t \geq 0, \tag{2.1}$$

where u is the initial surplus. In practice, the initial surplus is always non-negative, i.e. $u \geq 0$. But for the convenience of mathematical analysis, we allow u to take any real value hereafter; $c > 0$ is the annualized expected income rate; $\{J_k; k = 1, 2, \ldots\}$ are independent and identically distributed random variables supported on $\mathbb{R}\setminus\{0\}$ and they represent successive jump sizes; the total number of jumps of the surplus up to time t, denoted by N_t and independent of $\{J_k; k = 1, 2, \ldots\}$, is a Poisson process with intensity $\lambda > 0$; $\{W_t; t \geq 0\}$, independent of $\left\{\sum_{k=1}^{N_t} J_k; t \geq 0\right\}$, is a standard Brownian motion and $\sigma > 0$ is the volatility of the diffusion.

Furthermore, the jumps $\{J_k; k = 1, 2, \ldots\}$ are assumed to follow a mixture of a positive random variable following an arbitrary distribution P and the negative of an exponential random variable, i.e.

$$\mathbb{P}(J_k \leq x) = \begin{cases} \alpha + (1-\alpha)P(x), & x > 0, \\ \alpha e^{\theta x}, & x \leq 0, \end{cases}$$

where $\alpha \in (0, 1)$ represents the probability of an upward jump and $\theta > 0$ is the exponential parameter. We exclude the cases $\alpha = 0$ and $\alpha = 1$ as the surplus process $\{\tilde{U}_t; t \geq 0\}$ is reduced to either a spectrally negative or positive Lévy process, both of which have been studied extensively.

For the threshold dividend strategy with threshold level $b > 0$ and dividend rate $0 < \gamma \leq c$, let U_t be the retained surplus at time t after dividend payments. Thus, the retained surplus process in differential form can be written as

$$\mathrm{d}U_t = (c - \gamma \mathbb{I}(U_t > b))\,\mathrm{d}t - \mathrm{d}\sum_{k=1}^{N_t} J_k + \sigma \mathrm{d}W_t, \quad t > 0, \quad \text{and} \quad U_0 = u, \tag{2.2}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

As usual, the time of (ultimate) ruin is defined as $\tau \triangleq \inf\{t \geq 0 : U_t \leq 0\}$, where $\tau = 0$ if $u \leq 0$ and $\tau = \infty$ if ruin does not occur in finite time. Obviously, ruin can be caused either by oscillation (diffusion) or by a downward jump. The EDP function of Gerber and Shiu (1998) is defined by

$$\Psi(u; b) \triangleq \mathbb{E}[e^{-\delta\tau}\omega(U_\tau)\mathbb{I}(\tau < \infty)|U_0 = u], \tag{2.3}$$

where $\delta > 0$ is interpreted as the force of interest or the variable of a Laplace transform and ω is a penalty function that is non-negative and defined on $(-\infty, 0]$. It is well known that almost all the quantities of interest related to the time of ruin τ are special cases of the EDP function. Note that the penalty function ω is defined slightly differently from the usual definition that was originally introduced in Gerber and Shiu (1998).

In the next two sections, we present some probabilistic results for the surplus process defined in (2.1) with no initial surplus. These results will be used to obtain an analytical expression of the EDP function for the retained surplus process under the threshold dividend strategy in Section 5.

3. Wiener–Hopf factorization

In this section, we analyze the surplus process (2.1) with no initial surplus and provide explicit expressions for the distributions of its supremum and infimum processes killed at rate δ. For notational convenience, let

$$X_t \triangleq \tilde{U}_t - \tilde{U}_0, \tag{3.1}$$

where $\{\tilde{U}_t; t \geq 0\}$ is defined in (2.1). Obviously, it is a Lévy process with characteristic exponent

$$\frac{1}{t}\ln\left(\mathbb{E}\left[e^{sX_t}\right]\right) = cs + \frac{\sigma^2}{2}s^2 + \lambda\left\{(1-\alpha)[\hat{P}(s) - 1] + \frac{\alpha s}{\theta - s}\right\}, \quad t > 0, \quad \mathscr{R}s = 0,$$

where $\mathscr{R}s$ represents the real part of a complex number s and $\hat{P}(s) \triangleq \int_0^\infty e^{-sx} dP(x)$ is the Laplace–Stieltjes transform of the distribution function $P(x)$.

The right-hand side of the above equation as a function of s on the real line plays a key role in the analysis of the surplus process. Define

$$\phi(s) \triangleq cs + \frac{\sigma^2}{2}s^2 + \lambda\left\{(1-\alpha)[\hat{P}(s)-1] + \frac{\alpha s}{\theta - s}\right\} - \delta. \quad (3.2)$$

The equation

$$\phi(s) = 0 \quad (3.3)$$

is referred to as the *Lundberg fundamental equation* in the ruin theory context. When constrained on $\mathbb{R}_+$, the function $\phi(s)$ has only a pole: $s = \theta$. The fact that $-\infty = \phi(\theta+) < \phi(\theta-) = \infty$, together with $\lim_{s\to\infty}\phi(s) = \infty$ and $\phi(0) = -\delta < 0$, yields that the Lundberg fundamental equation (3.3) has at least two distinct positive roots ξ_1 and ξ_2 with

$$0 < \xi_1 < \theta < \xi_2 < \infty.$$

Let $D \triangleq \frac{\sigma^2}{2}$ and $\zeta \triangleq \xi_1 + \xi_2 - \theta + c/D > 0$. Define

$$h_1(x) \triangleq \zeta e^{-\zeta x} \quad \text{and}$$
$$h_2(x) \triangleq \frac{\lambda(1-\alpha)}{D\zeta}\left[\frac{\theta-\xi_1}{\xi_2-\xi_1}g_{\xi_1}(x) + \frac{\xi_2-\theta}{\xi_2-\xi_1}g_{\xi_2}(x)\right], \quad x > 0,$$

where $g_l(x) \triangleq \int_x^\infty e^{-l(y-x)}dP(y)$ for any $\mathscr{R}l \geq 0$, and $h(x) \triangleq \int_0^x h_1(x-y)h_2(y)dy$. Moreover, following Lin and Willmot (1999), we define

$$H(x) \triangleq \int_0^x h(y)dy \quad \text{and} \quad K(x) \triangleq q\sum_{j=0}^\infty H^{*j}(x). \quad (3.4)$$

In the above, $q \triangleq \frac{\theta\delta}{D\xi_1\xi_2\zeta}$ and $H^{*j}(x)$ is the j-fold convolution of $H(x)$ with itself, where the convolution of two distribution functions $F(x), x \geq 0$, and $G(x), x \geq 0$, is defined as $(F*G)(x) = \int_{0-}^x F(x-y)dG(y)$, $x \geq 0$. Thus, $H(x)$ is a defective distribution function while $K(x)$ is a compound geometric distribution function with probability mass q at 0.

The supremum process and the infimum process of $\{X_t; t \geq 0\}$ are given by

$$\overline{X}_t = \sup_{0\leq s\leq t} X_s \quad \text{and} \quad \underline{X}_t = \inf_{0\leq s\leq t} X_s.$$

Let $e(\delta)$ be an exponential random variable with mean $1/\delta$ that is independent of $\{X_t; t \geq 0\}$. By virtue of Bertoin (1996) and Kyprianou (2006), $X_{e(\delta)}$ has the following Wiener–Hopf factorization:

$$\mathbb{E}\left[e^{sX_{e(\delta)}}\right] = \mathbb{E}\left[e^{s\overline{X}_{e(\delta)}}\right]\mathbb{E}\left[e^{s\underline{X}_{e(\delta)}}\right] = -\frac{\delta}{\phi(s)}, \qquad \mathscr{R}s = 0, \quad (3.5)$$

where $\phi(s)$ is given in (3.2).

We now state the main result of this section, which generalizes the results of Proposition 3.2 in Chi (2010).

Proposition 3.1. *The probability density function (p.d.f.) of $\overline{X}_{e(\delta)}$ is given by*

$$f_{\overline{X}_{e(\delta)}}(x) = \frac{\xi_2(\theta-\xi_1)}{\theta(\xi_2-\xi_1)}\xi_1 e^{-\xi_1 x} + \frac{\xi_1(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}\xi_2 e^{-\xi_2 x}, \quad x > 0, \quad (3.6)$$

and the p.d.f. of $-\underline{X}_{e(\delta)}$ is represented by

$$f_{-\underline{X}_{e(\delta)}}(x) = \int_{0-}^x \zeta e^{-\zeta(x-y)}dK(y), \quad x > 0. \quad (3.7)$$

Proof. First, recall that Theorem 2.2 in Lewis and Mordecki (2008) demonstrates that

$$\phi^+(s) \triangleq \mathbb{E}\left[e^{s\overline{X}_{e(\delta)}}\right] = \frac{\xi_2(\theta-\xi_1)}{\theta(\xi_2-\xi_1)}\frac{\xi_1}{\xi_1-s} + \frac{\xi_1(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}\frac{\xi_2}{\xi_2-s}, \quad \text{for all } \mathscr{R}s = 0. \quad (3.8)$$

Thus, (3.6) is obtained by inverting (3.8).

Second, the Wiener–Hopf factorization (3.5) implies that

$$\phi^-(s) \triangleq \mathbb{E}\left[e^{s\underline{X}_{e(\delta)}}\right] = -\frac{\delta}{\phi(s)\mathbb{E}\left[e^{s\overline{X}_{e(\delta)}}\right]} = -\frac{\delta\theta}{\xi_1\xi_2}\times\frac{(\xi_1-s)(\xi_2-s)}{(\theta-s)\phi(s)}. \quad (3.9)$$

The following proof is similar to that of Proposition 3.2 in Chi (2010). Since ξ_1 is the positive root of the Lundberg fundamental equation (3.3), we have

$$\frac{\phi(s)}{s-\xi_1} = \frac{\phi(s)-\phi(\xi_1)}{s-\xi_1} = D(s+\xi_1) + c + \lambda(1-\alpha)\frac{\hat{P}(s)-\hat{P}(\xi_1)}{s-\xi_1} + \frac{\lambda\theta\alpha}{(\theta-s)(\theta-\xi_1)}. \quad (3.10)$$

Note that ξ_2 is another positive root of Eq. (3.3). Thus, $\frac{\phi(\xi_2)}{\xi_2-\xi_1} = 0$ implies that

$$\frac{\phi(s)}{(s-\xi_1)(s-\xi_2)} = \frac{\frac{\phi(s)}{s-\xi_1} - \frac{\phi(\xi_2)}{\xi_2-\xi_1}}{s-\xi_2} = D + \lambda(1-\alpha)\frac{\frac{\hat{P}(s)-\hat{P}(\xi_1)}{s-\xi_1} - \frac{\hat{P}(\xi_2)-\hat{P}(\xi_1)}{\xi_2-\xi_1}}{s-\xi_2} + \frac{\lambda\alpha\theta}{(\theta-s)(\theta-\xi_1)(\theta-\xi_2)}.$$

Multiplying both sides of the above equation by $s-\theta$, we have with straightforward algebra

$$\begin{aligned}\frac{\phi(s)(s-\theta)}{(s-\xi_1)(s-\xi_2)} &= D\left(s+\xi_1+\xi_2-\theta+\frac{c}{D}\right) + \lambda(1-\alpha) \\ &\times\frac{\hat{P}(s)-\hat{P}(\xi_1)}{s-\xi_1} + \lambda(1-\alpha)(\xi_2-\theta)\frac{\frac{\hat{P}(s)-\hat{P}(\xi_1)}{s-\xi_1} - \frac{\hat{P}(\xi_2)-\hat{P}(\xi_1)}{\xi_2-\xi_1}}{s-\xi_2} \\ &= D(s+\zeta)\left\{1 - \frac{\zeta}{s+\zeta}\frac{\lambda(1-\alpha)}{D\zeta}\left[-(\xi_2-\theta)\right.\right. \\ &\left.\left.\times\frac{\frac{\hat{P}(s)-\hat{P}(\xi_1)}{s-\xi_1} - \frac{\hat{P}(\xi_2)-\hat{P}(\xi_1)}{\xi_2-\xi_1}}{s-\xi_2} - \frac{\hat{P}(s)-\hat{P}(\xi_1)}{s-\xi_1}\right]\right\}, \quad (3.11)\end{aligned}$$

where the first equality is derived from (3.10) for the case $s = \xi_2$.

Let $\hat{h_2}(s) = \int_0^\infty e^{-sx}h_2(x)dx$ be the Laplace transform of function $h_2(x)$, and let $\hat{h_1}(s)$ and $\hat{h}(s)$ be defined similarly. Since

$$\frac{D\zeta}{\lambda(1-\alpha)}\hat{h_2}(s) = -(\xi_2-\theta)\frac{\frac{\hat{P}(s)-\hat{P}(\xi_1)}{s-\xi_1} - \frac{\hat{P}(\xi_2)-\hat{P}(\xi_1)}{\xi_2-\xi_1}}{s-\xi_2} - \frac{\hat{P}(s)-\hat{P}(\xi_1)}{s-\xi_1},$$

(3.11) can be rewritten as

$$\frac{\phi(s)(s-\theta)}{D(s-\xi_1)(s-\xi_2)(s+\zeta)} = 1 - \hat{h_1}(s)\times\hat{h_2}(s) = 1 - \hat{h}(s).$$

This equation and (3.9) together result in

$$\phi^-(s) = q\frac{\zeta}{s+\zeta}\{1-\hat{h}(s)\}^{-1} = q\hat{h_1}(s)\{1-\hat{h}(s)\}^{-1}, \qquad \mathscr{R}s = 0.$$

It is then easy to see that $\phi^-(-s)$ is the characteristic function of

$$q\int_{0-}^{x}\zeta e^{-\zeta(x-y)}d\sum_{j=0}^{\infty}H^{*j}(y) = \int_{0-}^{x}\zeta e^{-\zeta(x-y)}dK(y),$$

where $H(x)$ and $K(x)$ are defined in (3.4). The proof is therefore complete. □

With the help of Proposition 3.1, we will derive the joint distribution of the barrier hitting time and the corresponding overflow/deficit for $\{X_t; t \geq 0\}$ defined in (3.1) in the next section.

4. Joint distribution of barrier hitting time and overflow/deficit

We first focus on the case that there is one barrier. The joint distribution of the upper barrier hitting time and the overflow and the joint distribution of the lower barrier hitting time and the deficit for process $\{X_t; t \geq 0\}$ are obtained in Section 4.1. When there are two barriers: one upper and one lower, the joint distribution of the barrier hitting time and overflow/deficit is more complicated. This distribution is obtained in Section 4.2 by using the approach in Kadankova and Veraverbeke (2007) in which the two-sided exit problem for a compound Poisson process with exponential upward jumps is analyzed.

4.1. The case of one barrier

Denote the upper barrier hitting time and lower barrier hitting time by

$$\begin{aligned}&\tau_a^+ = \inf\{t \geq 0 : X_t > a\} \quad \text{and}\\ &\tau_a^- = \inf\{t \geq 0 : X_t < -a\}, \quad a \geq 0\end{aligned} \tag{4.1}$$

respectively. The corresponding overflow and deficit are

$$H_X(a) \triangleq X_{\tau_a^+} - a \quad \text{and} \quad L_X(a) \triangleq -X_{\tau_a^-} - a. \tag{4.2}$$

Proposition 4.1. *The two-dimensional Laplace transform of the upper barrier hitting time and the overflow is given by*

$$\mathbb{E}\left[e^{-\delta\tau_a^+ - sH_X(a)}\right] = \frac{\theta-\xi_1}{\xi_2-\xi_1}e^{-\xi_1 a}\left(1+\frac{\xi_2-\theta}{\theta+s}\right) + \frac{\xi_2-\theta}{\xi_2-\xi_1}e^{-\xi_2 a}\left(1+\frac{\xi_1-\theta}{\theta+s}\right). \tag{4.3}$$

Moreover, the discounted density of the overflow is

$$\mathbb{E}\left[e^{-\delta\tau_a^+}\mathbb{I}(H_X(a)\in dy)\right] = \left(\frac{\theta-\xi_1}{\xi_2-\xi_1}e^{-\xi_1 a} + \frac{\xi_2-\theta}{\xi_2-\xi_1}e^{-\xi_2 a}\right) \times \delta_0(y)dy + \frac{(\theta-\xi_1)(\xi_2-\theta)}{(\xi_2-\xi_1)}[e^{-\xi_1 a} - e^{-\xi_2 a}]e^{-\theta y}dy \tag{4.4}$$

for any $y \geq 0$, where $\delta_0(y)$ is a Dirac delta function centered at 0.

Proof. Recall that Theorem 1 in Bingham (1975) demonstrates that

$$\int_0^\infty e^{-la}\mathbb{E}\left[e^{-\delta\tau_a^+ - sH_X(a)}\right]da = \frac{1}{l-s}\left\{1-\frac{\phi^+(-l)}{\phi^+(-s)}\right\} \tag{4.5}$$

for $\delta > 0$, $l > 0$ and $\mathscr{R}s \geq 0$, where $\phi^+(s)$ is given in (3.8). Since

$$\int_0^\infty e^{-lx}\left(e^{sx}\int_x^\infty e^{-sy}f_{\overline{X}_{e(\delta)}}(y)dy\right)dx = \frac{\phi^+(-s)-\phi^+(-l)}{l-s}$$

(see Li and Garrido (2004)), (4.5) implies

$$\mathbb{E}\left[e^{-\delta\tau_a^+ - sH_X(a)}\right] = \frac{\int_a^\infty e^{-s(y-a)}f_{\overline{X}_{e(\delta)}}(y)dy}{\phi^+(-s)} = \frac{\theta-\xi_1}{\xi_2-\xi_1}e^{-\xi_1 a}\left(1+\frac{\xi_2-\theta}{\theta+s}\right) + \frac{\xi_2-\theta}{\xi_2-\xi_1}e^{-\xi_2 a}\left(1+\frac{\xi_1-\theta}{\theta+s}\right).$$

The second equality is due to (3.6) and (3.8). Inverting (4.3) immediately leads to (4.4). □

Remark 4.1. Taking $s = 0$ in (4.3), we have

$$\mathbb{E}\left[e^{-\delta\tau_t^+}\right] = \frac{\xi_2(\theta-\xi_1)}{\theta(\xi_2-\xi_1)}e^{-\xi_1 t} + \frac{\xi_1(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}e^{-\xi_2 t} \tag{4.6}$$

for any $t \in \mathbb{R}_+$. Different from the case for spectrally negative Lévy processes, formula (4.6) implies that $\{\tau_t^+; t \geq 0\}$ is not a subordinator anymore. Moreover, (4.4) indicates that the upper barrier hitting time τ_a^+ is dependent on the overflow $H_X(a)$. However, conditional on $H_X(a) > 0$, the quantity of the overflow $H_X(a)$ is exponentially distributed with parameter θ and independent of the stopping time τ_a^+, as expected.

We now consider the lower barrier hitting time and the deficit.

Proposition 4.2. *For any $a > 0$ and $y \geq 0$, the discounted density of the deficit is given by*

$$\mathbb{E}[e^{-\delta\tau_a^-}\mathbb{I}(L_X(a)\in dy)] = \frac{1}{q\zeta}f_{-\underline{X}_{e(\delta)}}(a)[\delta_0(y)-h(y)]dy + \frac{1}{q}\left[K'(a+y) - \int_0^y h(y-z)dK(a+z)\right]dy, \tag{4.7}$$

where $f_{-\underline{X}_{e(\delta)}}(x)$ is given in (3.7).

Proof. Since the proof is similar to that of Theorem 3.1 in Chi (2010), we omit the details. □

4.2. The case of two barriers

For $0 < u \leq b$, denote the barrier hitting time of $\{X_t; t \geq 0\}$ with lower barrier $-u$ and upper barrier $b-u$ (they are corresponding to the barriers 0 and b for the surplus process $\tilde{U}_t$) by

$$\tau_u^b \triangleq \inf\{t \geq 0 : u + X_t \notin [0, b]\}, \tag{4.8}$$

the first time the process $\{X_t; t \geq 0\}$ hits one of the barriers. Since $\tau_u^b = \min\{\tau_u^-, \tau_{b-u}^+\}$ in terms of the single barrier hitting times defined in (4.1), we may write the corresponding overflow/deficit as

$$\Lambda \triangleq H_X(b-u)\mathbb{I}(\tau_u^b = \tau_{b-u}^+) + L_X(u)\mathbb{I}(\tau_u^b = \tau_u^-). \tag{4.9}$$

Further, denote the inner product of functions f_1 and f_2 defined on $\mathbb{R}$ by

$$\langle f_1, f_2\rangle \triangleq \int_{-\infty}^{\infty} f_1(x)f_2(x)dx$$

and

$$\mathcal{L}_\beta(u) \triangleq \mathbb{E}\left[e^{-\delta\tau_u^- - \beta L_X(u)}\right] = \frac{\int_u^\infty e^{-\beta(y-u)}f_{-\underline{X}_{e(\delta)}}(y)dy}{\phi^-(\beta)}, \quad \beta > 0, \tag{4.10}$$

where $f_{-\underline{X}_{e(\delta)}}(y)$ and $\phi^-(s)$ are given by (3.7) and (3.9) respectively.

Moreover, define

$$\eta \triangleq \Pi_{j=1}^2(1-\langle\tilde{f}_j, \tilde{g}_j\rangle) - \langle\tilde{f}_1, \tilde{g}_2\rangle\langle\tilde{f}_2, \tilde{g}_1\rangle, \tag{4.11}$$

where

$$\begin{aligned}
\tilde{f}_1(x) &\triangleq \frac{\theta-\xi_1}{\xi_2-\xi_1}\mathrm{e}^{-\xi_1(b+x)}+\frac{\xi_2-\theta}{\xi_2-\xi_1}\mathrm{e}^{-\xi_2(b+x)};\\
\tilde{f}_2(x) &\triangleq \frac{(\theta-\xi_1)(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}(\mathrm{e}^{-\xi_1(b+x)}-\mathrm{e}^{-\xi_2(b+x)});\\
\tilde{g}_1(x)\mathrm{d}x &\triangleq \mathbb{E}[\mathrm{e}^{-\delta\tau_b^-}\mathbb{I}(L_X(b)\in\mathrm{d}x)];\\
\tilde{g}_2(x)\mathrm{d}x &\triangleq \mathbb{E}[\mathrm{e}^{-\delta\tau_{b+e(\theta)}^-}\mathbb{I}(L_X(b+e(\theta))\in\mathrm{d}x)]
\end{aligned}\tag{4.12}$$

for $x\geq 0$, and

$$\begin{aligned}
r_1(u) &\triangleq \frac{1}{\eta}\Bigg(\frac{\theta-\xi_1}{\xi_2-\xi_1}\\
&\quad+\frac{(\theta-\xi_1)(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}\mathrm{e}^{-\xi_2 b}\mathbb{E}[\mathcal{L}_{\xi_2}(e(\theta)+b)]\Bigg)\\
&\quad\times\upsilon_1(u)+\frac{1}{\eta}\Bigg(\frac{\xi_2-\theta}{\xi_2-\xi_1}-\frac{(\theta-\xi_1)(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}\mathrm{e}^{-\xi_1 b}\\
&\quad\times\mathbb{E}[\mathcal{L}_{\xi_1}(e(\theta)+b)]\Bigg)\upsilon_2(u),\\
r_2(u) &\triangleq \frac{(\theta-\xi_1)(\xi_2-\theta)}{\eta\theta(\xi_2-\xi_1)}\left\{\upsilon_1(u)[1-\mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(b)]\right.\\
&\quad\left.-\upsilon_2(u)[1-\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(b)]\right\},
\end{aligned}\tag{4.13}$$

where

$$\upsilon_j(u)\triangleq \mathrm{e}^{-\xi_j b}\left(\mathrm{e}^{\xi_j u}-\mathcal{L}_{\xi_j}(u)\right),\quad j=1,2.\tag{4.14}$$

We show in Appendix B that

$$r_1(b)=1,\qquad r_2(b)=0,\qquad r_1(0)=r_2(0)=0.\tag{4.15}$$

With the help of Propositions 4.1 and 4.2, we obtain the main result of this subsection for the joint distribution of the barrier hitting time defined in (4.8) and its overflow/deficit in (4.9).

Proposition 4.3. *For any $0\leq u\leq b$ and $y\geq 0$, we have*

$$\mathbb{E}[\mathrm{e}^{-\delta\tau_u^b}\mathbb{I}(\Lambda\in\mathrm{d}y,\tau_u^b=\tau_{b-u}^+)]=r_1(u)\delta_0(y)\mathrm{d}y+r_2(u)\theta\mathrm{e}^{-\theta y}\mathrm{d}y\tag{4.16}$$

and

$$\begin{aligned}
&\mathbb{E}[\mathrm{e}^{-\delta\tau_u^b}\mathbb{I}(\Lambda\in\mathrm{d}y,\tau_u^b=\tau_u^-)]\\
&\quad=\mathbb{E}[\mathrm{e}^{-\delta\tau_u^-}\mathbb{I}(L_X(u)\in\mathrm{d}y)]-\left(r_1(u)\tilde{g}_1(y)+r_2(u)\tilde{g}_2(y)\right)\mathrm{d}y.
\end{aligned}\tag{4.17}$$

Proof. The proof is lengthy and is given in Appendix A. □

Remark 4.2. Functions $r_1(u)$ and $r_2(u)$ have the following probabilistic interpretation:

$$\begin{aligned}
&\mathbb{E}[\mathrm{e}^{-\delta\tau_u^b}\mathbb{I}(\Lambda=0,\tau_u^b=\tau_{b-u}^+)]=r_1(u)\quad\text{and}\\
&\mathbb{E}[\mathrm{e}^{-\delta\tau_u^b}\mathbb{I}(\Lambda>0,\tau_u^b=\tau_{b-u}^+)]=r_2(u)
\end{aligned}\tag{4.18}$$

for any $0\leq u\leq b$.

We now have the required probabilistic results to carry out the analysis of the Gerber–Shiu EDP function for the generalized insurance risk model under the threshold dividend strategy described in (2.2).

5. Analysis of the EDP function

In this section, we derive an analytical expression for the EDP function of the risk model (2.2) with and without dividends.

If no dividends are paid out, it is equivalent to setting the threshold level $b=\infty$. In this case, the insurer's surplus is reduced to $\tilde{U}_t$ and the ruin time is denoted by $\tilde{\tau}\triangleq\inf\{t\geq 0:\tilde{U}_t<0\}$. By virtue of Proposition 4.2, we obtain an analytical expression for the EDP function in the following theorem.

Theorem 5.1. *Denote*

$$\begin{aligned}
\Omega(u) &\triangleq \omega(0)\mathrm{e}^{-\zeta u}+\int_u^\infty\omega(u-x)h(x)\mathrm{d}x\\
&\quad-\mathrm{e}^{-\zeta u}\int_0^\infty\omega(-x)h(x)\mathrm{d}x,\quad u\geq 0.
\end{aligned}$$

Then the EDP function (2.3) for the surplus $\tilde{U}_t$ without dividends can be expressed as

$$\Phi(u)=\frac{1}{q}\left(\Omega*K\right)(u),\quad u\geq 0.\tag{5.1}$$

Proof. When $u=0$, the above result is trivial. For $u>0$, we have

$$\mathbb{E}[\mathrm{e}^{-\delta\tilde{\tau}}\mathbb{I}(-\tilde{U}_{\tilde{\tau}}\in\mathrm{d}y)|\tilde{U}_0=u]=\mathbb{E}[\mathrm{e}^{-\delta\tau_u^-}\mathbb{I}(L_X(u)\in\mathrm{d}y)],\quad y\geq 0.$$

Thus, (5.1) follows from (4.7) by taking the same argument as that of Theorem 3.1 in Chi (2010). The proof is complete. □

For $u<0$, it is obvious that the EDP function $\Phi(u)=\omega(u)$ and hence the EDP function is defined for all u. We now turn to the threshold dividend strategy with $b<\infty$ being the threshold level. Recall that dividends are paid out at rate γ as long as the insurer's retained surplus is above b. In this case, the insurer's retained surplus process $\{U_t;t\geq 0\}$ satisfies

$$\mathrm{d}U_t=\tilde{c}\mathrm{d}t-\mathrm{d}\sum_{k=1}^{N_t}J_k+\sigma\mathrm{d}W_t,\quad U_t>b,$$

where $\tilde{c}\triangleq c-\gamma\geq 0$.

The corresponding Lundberg fundamental equation is

$$\tilde{c}s+\frac{\sigma^2}{2}s^2+\lambda\left((1-\alpha)(\hat{P}(s)-1)+\alpha\times\frac{s}{\theta-s}\right)-\delta=0,$$

and it has two distinct positive roots $\{\tilde{\xi}_1,\tilde{\xi}_2\}$ satisfying $0<\tilde{\xi}_1<\theta<\tilde{\xi}_2<\infty$.

Similar to Section 3, we introduce some auxiliary functions. Denote

$$\tilde{\zeta}\triangleq\tilde{\xi}_1+\tilde{\xi}_2-\theta+\tilde{c}/D,\qquad \tilde{q}\triangleq\theta\delta/(D\tilde{\xi}_1\tilde{\xi}_2\tilde{\zeta})$$

and define

$$\begin{aligned}
\tilde{h}_1(x) &\triangleq \tilde{\zeta}\mathrm{e}^{-\tilde{\zeta}x},\\
\tilde{h}_2(x) &\triangleq \frac{\lambda(1-\alpha)}{D\tilde{\zeta}}\left\{\frac{\theta-\tilde{\xi}_1}{\tilde{\xi}_2-\tilde{\xi}_1}g_{\tilde{\xi}_1}(x)+\frac{\tilde{\xi}_2-\theta}{\tilde{\xi}_2-\tilde{\xi}_1}g_{\tilde{\xi}_2}(x)\right\},\quad x>0,
\end{aligned}$$

and $\tilde{h}(x)=\int_0^x\tilde{h}_1(x-y)\tilde{h}_2(y)\mathrm{d}y$. Also,

$$\tilde{H}(x)\triangleq\int_0^x\tilde{h}(y)\mathrm{d}y\quad\text{and}\quad\tilde{K}(x)\triangleq\tilde{q}\sum_{n=0}^\infty\tilde{H}^{*n}(x).\tag{5.2}$$

Moreover, define

$$\begin{aligned}
\tilde{\Omega}(u) &= \int_{b+u}^\infty\omega(b+u-y)\tilde{h}(y)\mathrm{d}y-\mathrm{e}^{-\tilde{\zeta}u}\int_b^\infty\omega(b-y)\tilde{h}(y)\mathrm{d}y,\\
\tilde{F}(u) &= \int_u^{b+u}f(b+u-y)\tilde{h}(y)\mathrm{d}y-\mathrm{e}^{-\tilde{\zeta}u}\int_0^b f(b-y)\tilde{h}(y)\mathrm{d}y,
\end{aligned}$$

and

$$\mathscr{E}_f(u) = \frac{1}{\tilde{q}}\left(\tilde{F} * \tilde{K}\right)(u), \quad u \geq 0.$$

The last expression may be viewed as a function whose domain is the set of functions f defined on $[0, b]$. We now have the following proposition.

Proposition 5.1. *The EDP function $\Psi(u; b)$ defined in (2.3) satisfies the piecewise integral equation*

$$\Psi(u; b) = \begin{cases} \Psi(b; b)I(u-b) + \mathscr{E}_\Psi(u-b) + R(u-b), & u > b, \\ \Phi(u) + r_1(u)(\Psi(b; b) - \Phi(b)) \\ \quad + r_2(u)(\mathbb{E}[\Psi(e(\theta)+b; b)] \\ \quad - \mathbb{E}[\Phi(e(\theta)+b)]), & 0 \leq u \leq b, \\ \omega(u), & u < 0, \end{cases} \tag{5.3}$$

where $\Phi(u)$ is given in (5.1), and

$$I(u) \triangleq \frac{1}{\tilde{q}}\int_{0-}^{u} \mathrm{e}^{-\tilde{\zeta}(u-y)}\mathrm{d}\tilde{K}(y), \qquad R(u) \triangleq \frac{1}{\tilde{q}}\left(\tilde{\Omega} * \tilde{K}\right)(u), \quad u \geq 0.$$

Proof. It is obvious that $\Psi(u; b) = \omega(u)$ for $u \leq 0$. Meanwhile, for any $0 \leq u \leq b$, there are two possibilities: the ruin occurs before the insurer's retained surplus has been over the threshold level b or it occurs after it reaches the threshold level. Thus, we have

$$\begin{aligned} \Psi(u; b) &= \mathbb{E}[\mathrm{e}^{-\delta\tau}\omega(U_\tau)\mathbb{I}(\tau < \tau^+_{b-u} \leq \infty)|U_0 = u] \\ &\quad + \mathbb{E}[\mathrm{e}^{-\delta\tau}\omega(U_\tau)\mathbb{I}(\tau^+_{b-u} < \tau < \infty)|U_0 = u] \\ &= \mathbb{E}[\mathrm{e}^{-\delta\tau^-_u}\omega(-L_X(u))\mathbb{I}(\tau^b_u = \tau^-_u < \infty)] \\ &\quad + \mathbb{E}\left[\mathbb{E}[\mathrm{e}^{-\delta\tau}\omega(U_\tau)\mathbb{I}(\tau^+_{b-u} < \tau < \infty)|\mathscr{F}_{\tau^+_{b-u}}]|U_0 = u\right] \\ &= \Phi(u) - r_1(u)\Phi(b) - r_2(u)\mathbb{E}[\Phi(e(\theta)+b)] \\ &\quad + \mathbb{E}[\mathrm{e}^{-\delta\tau^+_{b-u}}\Psi(b + H_X(b-u); b)\mathbb{I}(\tau^+_{b-u} = \tau^b_u < \infty)] \\ &= \Phi(u) + r_1(u)(\Psi(b; b) - \Phi(b)) \\ &\quad + r_2(u)(\mathbb{E}[\Psi(e(\theta)+b; b)] - \mathbb{E}[\Phi(e(\theta)+b)]), \end{aligned} \tag{5.4}$$

where $\{\mathscr{F}_t\}_{t\geq 0}$ is the augmentation of the natural filtration of $\{U_t; t \geq 0\}$. The third equality is derived from (4.17) and the strong Markov property of $\{U_t; t \geq 0\}$, and the last equality is obtained from (4.16).

When $u > b$, we localize $\{U_t; t \geq 0\}$ through the stopping time $\mathcal{T} \triangleq \inf\{t \geq 0 : U_t \leq b\} \leq \tau$. In this case, we can take it as a perturbed compound Poisson risk model with premium rate $\tilde{c}$, initial reserve $u - b > 0$ and penalty function $\Psi(b + y; b)$ for any $y \in \mathbb{R}_-$. Therefore, (5.1) implies

$$\Psi(u; b) = \frac{1}{\tilde{q}}\left(\tilde{\Upsilon} * \tilde{K}\right)(u-b), \tag{5.5}$$

where

$$\begin{aligned} \tilde{\Upsilon}(u) &= \Psi(b; b)\mathrm{e}^{-\tilde{\zeta}u} + \int_u^\infty \Psi(b+u-y; b)\tilde{h}(y)\mathrm{d}y \\ &\quad - \mathrm{e}^{-\tilde{\zeta}u}\int_0^\infty \Psi(b-y; b)\tilde{h}(y)\mathrm{d}y. \end{aligned}$$

Eq. (5.5) is the same as the first equation in (5.3) since $\Psi(u; b) = \omega(u)$ for $u \leq 0$. This completes the proof. □

Before solving the integral equation (5.3) for $u \geq 0$, we again need to introduce some auxiliary functions. Let

$$\mathscr{S}(x) \triangleq R(x) - \Phi(b)\mathscr{E}_{r_1}(x) - \mathbb{E}[\Phi(b + e(\theta))]\mathscr{E}_{r_2}(x) + \mathscr{E}_\Phi(x)$$

and

$$\begin{aligned} w_1(x) &\triangleq I(x) + \mathscr{E}_{r_1}(x) + \mathscr{E}_{r_2}(x)\frac{\mathbb{E}[I(e(\theta))] + \mathbb{E}[\mathscr{E}_{r_1}(e(\theta))]}{1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]}; \\ w_2(x) &\triangleq r_1(x) + r_2(x)\frac{\mathbb{E}[I(e(\theta))] + \mathbb{E}[\mathscr{E}_{r_1}(e(\theta))]}{1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]}; \\ J(x) &\triangleq \Phi(x) - r_1(x)\Phi(b) + r_2(x)\left(\frac{\mathbb{E}[\mathscr{S}(e(\theta))]}{1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]}\right. \\ &\quad \left. - \mathbb{E}[\Phi(e(\theta)+b)]\right). \end{aligned}$$

Theorem 5.2. *For $u \geq 0$, the EDP function defined in (2.3) can be expressed explicitly as*

$$\Psi(u; b) = \begin{cases} \kappa w_1(u-b) + \dfrac{\mathbb{E}[\mathscr{S}(e(\theta))]}{1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]} \\ \quad \times \mathscr{E}_{r_2}(u-b) + \mathscr{S}(u-b), & u > b, \\ \kappa w_2(u) + J(u), & 0 \leq u \leq b, \end{cases} \tag{5.6}$$

where

$$\begin{aligned} \kappa &= \Psi(b; b) = \frac{1}{w_2'(b) - w_1'(0)} \\ &\quad \times \left(\frac{\mathbb{E}[\mathscr{S}(e(\theta))]}{1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]}\mathscr{E}'_{r_2}(0) + \mathscr{S}'(0) - J'(b)\right). \end{aligned} \tag{5.7}$$

Proof. It follows from (5.4) that

$$\begin{aligned} \mathscr{E}_\Psi(x) &= \mathscr{E}_\Phi(x) + \mathscr{E}_{r_1}(x)(\Psi(b; b) - \Phi(b)) \\ &\quad + \mathscr{E}_{r_2}(x)(\mathbb{E}[\Psi(e(\theta)+b; b)] - \mathbb{E}[\Phi(e(\theta)+b)]). \end{aligned}$$

Plugging the above equation into (5.3) yields

$$\begin{aligned} \Psi(u+b; b) &= \Psi(b; b)(I(u) + \mathscr{E}_{r_1}(u)) \\ &\quad + \mathbb{E}[\Psi(b + e(\theta); b)]\mathscr{E}_{r_2}(u) + \mathscr{S}(u), \quad u \geq 0. \end{aligned} \tag{5.8}$$

Thus, randomizing the constant u by $e(\theta)$ and taking the expectation of both sides of the above equation, we have

$$\begin{aligned} &\left(1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]\right)\mathbb{E}[\Psi(b + e(\theta); b)] \\ &= \Psi(b; b)\left(\mathbb{E}[I(e(\theta))] + \mathbb{E}[\mathscr{E}_{r_1}(e(\theta))]\right) + \mathbb{E}[\mathscr{S}(e(\theta))]. \end{aligned} \tag{5.9}$$

We now show that $1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))] > 0$. Since (4.18) implies that

$$0 \leq r_2(u) = \mathbb{E}[\mathrm{e}^{-\delta\tau^b_u}\mathbb{I}(\Lambda > 0, \tau^b_u = \tau^+_{b-u})] \leq 1, \quad 0 \leq u \leq b,$$

we have

$$\begin{aligned} \mathscr{E}_{r_2}(x) &\leq \frac{1}{\tilde{q}}\left((\tilde{H}_b - \tilde{H}) * \tilde{K}\right)(x) \\ &= \frac{1}{\tilde{q}}\left(\left(\tilde{H}_b * \tilde{K}\right)(x) - \tilde{q}\sum_{n=1}^{\infty}\tilde{H}^{*n}(x)\right) \\ &= \frac{1}{\tilde{q}}\left(\left(\tilde{H}_b * \tilde{K}\right)(x) - \tilde{K}(x)\right) + 1 < 1, \end{aligned}$$

where $\tilde{H}_b(x) = \tilde{H}(b + x)$.

Thus, (5.9) can be rewritten as

$$\begin{aligned} \mathbb{E}[\Psi(b + e(\theta); b)] &= \frac{1}{1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]}(\Psi(b; b)(\mathbb{E}[I(e(\theta))] \\ &\quad + \mathbb{E}[\mathscr{E}_{r_1}(e(\theta))]) + \mathbb{E}[\mathscr{S}(e(\theta))]). \end{aligned}$$

Plugging the above equation into (5.4) and (5.8), we have

$$\begin{cases} \Psi(u+b;b) = \Psi(b;b)w_1(u) \\ \quad + \dfrac{\mathbb{E}[\mathscr{I}(e(\theta))]}{1-\mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]}\mathscr{E}_{r_2}(u) + \mathscr{I}(u), \quad u \geq 0, \\ \Psi(u;b) = \Psi(b;b)w_2(u) + J(u), \quad 0 \leq u \leq b. \end{cases} \tag{5.10}$$

To eliminate $\Psi(b;b)$ from the above equation, we need to introduce another boundary condition. By using a similar argument to that in Wan (2007), we obtain the first-order boundary condition: $\Psi'(b-;b) = \Psi'(b+;b)$. Thus, (5.7) follows from (5.10) directly. The proof is now complete. □

Note that in Theorem 5.2, the EDP function is expressed in terms of functions $\Phi(u)$, $\mathbb{E}[\Phi(e(\theta)+b)]$, $\{w_j(x); j = 1, 2\}$ and $\{r_k(x); k = 1, 2\}$, which in turn depend on the compound geometric distributions $K(u)$ and $\tilde{K}(u)$ defined in (3.4) and (5.2) respectively. When the distribution of the downward jumps is of phase-type, $K(u)$ and $\tilde{K}(u)$ can be written as a compact matrix-type form as shown in Chi (2010). Further, Rolski et al. (1999) demonstrate that the family of phase-type distributions is dense in the class of all distributions defined on $[0, \infty)$ (see also Tijms (1994)). Therefore, in the next section, we consider the distribution of the downward jumps to be phase-type. We will obtain an explicit expression of the EDP function in this case.

6. Phase-type downward jumps

In this section, we begin with a brief introduction of the phase-type distributions. Let $\{\mathscr{M}_t; t \geq 0\}$ be a continuous-time Markov chain with finite state space $\{1, 2, \ldots, k, \Xi\}$ where Ξ is the only absorption state and all other states are transient. The intensity matrix of such a process is written by

$$\mathbf{Q} = \begin{pmatrix} \mathbf{A} & \mathbf{a} \\ \mathbf{0}^t & 0 \end{pmatrix},$$

where $\mathbf{a} = -\mathbf{A}\mathbf{1}$ and $\mathbf{1} = (1, 1, \ldots, 1)^t$ is a column vector of 1's.

Let the absorbing time be $\chi \triangleq \inf\{t \geq 0 : \mathscr{M}_t = \Xi\}$ and the initial distribution be $(\boldsymbol{\mu}, \mu_\Delta)$, where $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_k)$ is assigned for states $1, \ldots, k$ and the probability $\mu_\Delta = 1 - \boldsymbol{\mu}\mathbf{1}$ is distributed for the absorption state, then the distribution function of χ is given by

$$F_\chi(t) = 1 - \mathbb{P}(\chi > t) = 1 - \boldsymbol{\mu}e^{\mathbf{A}t}\mathbf{1}, \quad t \geq 0.$$

The distribution of χ is completely determined by $\boldsymbol{\mu}$ and sub-intensity matrix $\mathbf{A}$. χ is said to be phase-type distributed and its distribution function is represented by $F_\chi \sim PH(\boldsymbol{\mu}, \mathbf{A})$.

Denote the L_1 norm of $\boldsymbol{\mu}$ by $\|\boldsymbol{\mu}\| = \sum_{j=1}^k \mu_j$. If $\|\boldsymbol{\mu}\| = 1$, F_χ is a continuous distribution function; otherwise, F_χ is a zero-modified probability distribution function, then the corresponding p.d.f. is

$$f_\chi(t) = \begin{cases} \boldsymbol{\mu}e^{\mathbf{A}t}\mathbf{a}, & t > 0, \\ (1 - \|\boldsymbol{\mu}\|)\delta_0(t), & t = 0. \end{cases}$$

In some situations, we need to extend the definition of a phase-type distribution such that the distribution is defective. In this case, a defective phase-type distribution has a p.d.f. $f_\chi(t) = \boldsymbol{\mu}e^{\mathbf{A}t}\mathbf{a}$ for $t > 0$ and $F_\chi(\infty) = \|\boldsymbol{\mu}\| < 1$.

In the following, we will derive an explicit expression for (5.6) step by step.

Lemma 6.1. *Let the downward jumps be phase-type distributed with representation $P \sim PH(\boldsymbol{\mu}, \mathbf{A})$, then $H(x)$ defined in (3.4) is a defective phase-type distribution function with representation $PH(\boldsymbol{\vartheta}, \mathbf{B})$, where*

$$\boldsymbol{\vartheta} = \left(\frac{\lambda(1-\alpha)}{D\zeta}\boldsymbol{\mu}(\theta\mathbf{I} - \mathbf{A})(\xi_1\mathbf{I} - \mathbf{A})^{-1}(\xi_2\mathbf{I} - \mathbf{A})^{-1}, 0\right),$$

$$\mathbf{B} = \begin{pmatrix} \mathbf{A} & \mathbf{a} \\ \mathbf{0}^t & -\zeta \end{pmatrix}.$$

In this case, $K(x)$ is a zero-modified phase-type distribution function with representation $PH(\boldsymbol{\vartheta}, \mathbf{V})$, where $\mathbf{V} = \mathbf{B} + \mathbf{b}\boldsymbol{\vartheta}$ is a non-singular sub-intensity matrix and $\mathbf{b} = -\mathbf{B}\mathbf{1}$.

Moreover, $\tilde{H}(x)$ defined in (5.2) is a defective phase-type distribution function with representation $PH(\tilde{\boldsymbol{\vartheta}}, \tilde{\mathbf{B}})$, where

$$\tilde{\boldsymbol{\vartheta}} = \left(\frac{\lambda(1-\alpha)}{D\tilde{\zeta}}\boldsymbol{\mu}(\theta\mathbf{I} - \mathbf{A})(\tilde{\xi}_1\mathbf{I} - \mathbf{A})^{-1}(\tilde{\xi}_2\mathbf{I} - \mathbf{A})^{-1}, 0\right),$$

$$\tilde{\mathbf{B}} = \begin{pmatrix} \mathbf{A} & \mathbf{a} \\ \mathbf{0}^t & -\tilde{\zeta} \end{pmatrix},$$

and $\tilde{K}(x)$ is a zero-modified phase-type distribution function with representation $PH(\tilde{\boldsymbol{\vartheta}}, \tilde{\mathbf{V}})$, where $\tilde{\mathbf{V}} = \tilde{\mathbf{B}} + \tilde{\mathbf{b}}\tilde{\boldsymbol{\vartheta}}$ is a non-singular sub-intensity matrix and $\tilde{\mathbf{b}} = -\tilde{\mathbf{B}}\mathbf{1}$.

Proof. Since the proof is same as that of Lemma 3.1 in Chi (2010), thus it is omitted. □

Denote

$$\boldsymbol{\alpha} \triangleq (1, \mathbf{0}^t), \qquad \mathbf{T} \triangleq \begin{pmatrix} -\zeta & \zeta\boldsymbol{\vartheta} \\ \mathbf{0} & \mathbf{V} \end{pmatrix} \quad \text{and} \quad \tilde{\mathbf{T}} \triangleq \begin{pmatrix} -\tilde{\zeta} & \tilde{\zeta}\tilde{\boldsymbol{\vartheta}} \\ \mathbf{0} & \tilde{\mathbf{V}} \end{pmatrix}$$

and

$$\omega_1 \triangleq \omega(0) - \boldsymbol{\nu}_\omega\mathbf{1}, \qquad \boldsymbol{\nu}_\omega \triangleq -\int_0^\infty \omega(-l)\boldsymbol{\vartheta}e^{\mathbf{B}l}\mathbf{B}\mathrm{d}l,$$

$$\boldsymbol{\nu}_b \triangleq -\int_0^\infty \omega(-l)\tilde{\boldsymbol{\vartheta}}e^{\tilde{\mathbf{B}}(b+l)}\tilde{\mathbf{B}}\mathrm{d}l,$$

where $\omega(x)$ is the penalty function.

Proposition 6.1. *Let the downward jumps be phase-type distributed with representation $P \sim PH(\boldsymbol{\mu}, \mathbf{A})$, then $f_{-\underline{X}_{e(\delta)}}(x)$ given in (3.7) can be simplified to*

$$f_{-\underline{X}_{e(\delta)}}(x) = \boldsymbol{\alpha}e^{\mathbf{T}x}\mathbf{t}, \quad x > 0, \tag{6.1}$$

where $\mathbf{t} = -\mathbf{T}\mathbf{1}$. In this case, the EDP function without dividend payout given in (5.1) is expressed as

$$\Phi(u) = \frac{1}{q}\left[(\boldsymbol{\nu}_\omega - \omega(0)\boldsymbol{\vartheta})e^{\mathbf{V}u}\mathbf{1} + \omega_1\boldsymbol{\alpha}e^{\mathbf{T}u}\mathbf{1}\right], \quad u \geq 0, \tag{6.2}$$

and hence

$$\mathbb{E}[\Phi(e(\theta)+b)] = \frac{\theta}{q}\left[(\boldsymbol{\nu}_\omega - \omega(0)\boldsymbol{\vartheta})(\theta\mathbf{I} - \mathbf{V})^{-1}e^{\mathbf{V}b}\mathbf{1} + \omega_1\boldsymbol{\alpha}(\theta\mathbf{I} - \mathbf{T})^{-1}e^{\mathbf{T}b}\mathbf{1}\right].$$

Proof. If $P \sim PH(\boldsymbol{\mu}, \mathbf{A})$, Lemma 6.1 shows that $K(x)$ is a zero-modified phase-type distribution function with representation $PH(\boldsymbol{\vartheta}, \mathbf{V})$, then we have

$$f_{-\underline{X}_{e(\delta)}}(x) = \int_{0-}^{x} \zeta e^{-\zeta(x-y)}\mathrm{d}K(y) = \boldsymbol{\alpha}e^{\mathbf{T}x}\mathbf{t}, \quad x > 0$$

according to Lemma 2.1 in Ren (2005). Moreover, (6.2) can be obtained by taking the same analysis as that of Theorem 3.3 in Chi (2010). □

An interesting special case is when the penalty function $\omega(x) = 1$ for $x \leq 0$. This corresponds to the Laplace transform of the time of ruin. In this case, we have $\boldsymbol{\nu}_\omega = \boldsymbol{\vartheta}$, $\omega_1 = q$ and expression (6.2) is simplified to

$$\Phi(u) = \mathbb{E}[e^{-\delta\tilde{\tau}}\mathbb{I}(\tilde{\tau} < \infty)|\tilde{U}_0 = u] = \boldsymbol{\alpha}e^{\mathbf{T}u}\mathbf{1}, \quad u \geq 0.$$

Denote now

$$\boldsymbol{\alpha}_f \triangleq -\tilde{\boldsymbol{\vartheta}}\int_0^b f(b-z)e^{\tilde{\mathbf{B}}z}\tilde{\mathbf{B}}\mathrm{d}z$$

for arbitrary function f defined on $\mathbb{R}$, then we have the following proposition.

Proposition 6.2. *When $P \sim PH(\boldsymbol{\mu}, \mathbf{A})$, the functions $R(x)$, $\mathscr{E}_f(x)$ and $I(x)$ can be expressed as*

$$R(x) = \frac{1}{\tilde{q}}\left(\boldsymbol{\nu}_b e^{\tilde{\mathbf{V}}x}\mathbf{1} - \boldsymbol{\nu}_b \mathbf{1}\boldsymbol{\alpha} e^{\tilde{\mathbf{T}}x}\mathbf{1}\right),$$
$$\mathscr{E}_f(x) = \frac{1}{\tilde{q}}\left(\boldsymbol{\alpha}_f e^{\tilde{\mathbf{V}}x}\mathbf{1} - \boldsymbol{\alpha}_f \mathbf{1}\boldsymbol{\alpha} e^{\tilde{\mathbf{T}}x}\mathbf{1}\right),$$
$$I(x) = \frac{1}{\tilde{q}}\left(\boldsymbol{\alpha} e^{\tilde{\mathbf{T}}x}\mathbf{1} - \tilde{\boldsymbol{\vartheta}} e^{\tilde{\mathbf{V}}x}\mathbf{1}\right), \quad x \geq 0.$$

Proof. When $P \sim PH(\boldsymbol{\mu}, \mathbf{A})$, Lemma 6.1 shows that $\tilde{H}(x)$ and $\tilde{K}(x)$ defined in (5.2) are phase-type distribution functions, then $\tilde{\Omega}(x)$ can be simplified to

$$\begin{aligned}\tilde{\Omega}(x) &= \int_{b+x}^{\infty} \omega(b+x-y)\tilde{\boldsymbol{\vartheta}} e^{\tilde{\mathbf{B}}y}\tilde{\mathbf{b}}\mathrm{d}y - e^{-\tilde{\xi}x}\int_b^{\infty}\omega(b-y)\tilde{\boldsymbol{\vartheta}} e^{\tilde{\mathbf{B}}y}\tilde{\mathbf{b}}\mathrm{d}y \\ &= \boldsymbol{\nu}_b e^{\tilde{\mathbf{B}}x}\mathbf{1} - \boldsymbol{\nu}_b \mathbf{1} e^{-\tilde{\xi}x} \\ &= \boldsymbol{\nu}_b \mathbf{1}(1 - e^{-\tilde{\xi}x}) - \left(\boldsymbol{\nu}_b \mathbf{1} - \boldsymbol{\nu}_b e^{\tilde{\mathbf{B}}x}\mathbf{1}\right).\end{aligned}$$

Thus $R(x)$ defined in Proposition 5.1 can be rewritten as

$$\begin{aligned}R(x) &= \frac{1}{\tilde{q}}\left(\boldsymbol{\nu}_b \mathbf{1}(1 - e^{-\tilde{\xi}x}) * \tilde{K}(x) - \left(\boldsymbol{\nu}_b \mathbf{1} - \boldsymbol{\nu}_b e^{\tilde{\mathbf{B}}x}\mathbf{1}\right) * \tilde{K}(x)\right) \\ &= \frac{1}{\tilde{q}}\left(\boldsymbol{\nu}_b \mathbf{1}(1 - \boldsymbol{\alpha} e^{\tilde{\mathbf{T}}x}\mathbf{1}) - \left(\boldsymbol{\nu}_b \mathbf{1} - \boldsymbol{\nu}_b e^{\tilde{\mathbf{V}}x}\mathbf{1}\right)\right) \\ &= \frac{1}{\tilde{q}}\left(\boldsymbol{\nu}_b e^{\tilde{\mathbf{V}}x}\mathbf{1} - \boldsymbol{\nu}_b \mathbf{1}\boldsymbol{\alpha} e^{\tilde{\mathbf{T}}x}\mathbf{1}\right),\end{aligned}$$

where the second equality is obtained from Lemmas 2.1 and 2.5 in Ren (2005).

Similar arguments apply to $\mathscr{E}_f(x)$ and $I(x)$. □

The above proposition allows us to calculate the expectations:

$$\begin{aligned}\mathbb{E}[\mathscr{E}_{r_j}(e(\theta))] &= \frac{\theta}{\tilde{q}}\left(\boldsymbol{\alpha}_{r_j}(\theta\mathbf{I} - \tilde{\mathbf{V}})^{-1}\mathbf{1} - \boldsymbol{\alpha}_{r_j}\mathbf{1}\boldsymbol{\alpha}(\theta\mathbf{I} - \tilde{\mathbf{T}})^{-1}\mathbf{1}\right), \\ &\quad j = 1, 2; \\ \mathbb{E}[I(e(\theta))] &= \frac{\theta}{\tilde{q}}\left(\boldsymbol{\alpha}(\theta\mathbf{I} - \tilde{\mathbf{T}})^{-1}\mathbf{1} - \tilde{\boldsymbol{\vartheta}}(\theta\mathbf{I} - \tilde{\mathbf{V}})^{-1}\mathbf{1}\right); \\ \mathbb{E}[\mathscr{S}(e(\theta))] &= \frac{\theta}{\tilde{q}}\left\{-\boldsymbol{\rho}_2 \mathbf{1}\boldsymbol{\alpha}(\theta\mathbf{I} - \tilde{\mathbf{T}})^{-1}\mathbf{1} + \boldsymbol{\rho}_2(\theta\mathbf{I} - \tilde{\mathbf{V}})^{-1}\mathbf{1}\right\},\end{aligned} \tag{6.3}$$

where

$$\boldsymbol{\rho}_2 = \boldsymbol{\nu}_b - \Phi(b)\boldsymbol{\alpha}_{r_1} - \mathbb{E}[\Phi(e(\theta) + b)]\boldsymbol{\alpha}_{r_2} + \boldsymbol{\alpha}_{\Phi}. \tag{6.4}$$

We now state the main result of this section: an explicit matrix representation of the EDP function $\Psi(u; b)$ for $u \geq 0$ when the downward jumps are phase-type distributed.

Theorem 6.1. *When the downward jumps are phase-type distributed with representation $P \sim PH(\boldsymbol{\mu}, \mathbf{A})$, we have*

$$\Psi(u; b) = \frac{1}{\tilde{q}}\left((\kappa\tilde{q} - \boldsymbol{\rho}\mathbf{1})\boldsymbol{\alpha} e^{\tilde{\mathbf{T}}(u-b)}\mathbf{1} + \boldsymbol{\rho} e^{\tilde{\mathbf{V}}(u-b)}\mathbf{1}\right) \tag{6.5}$$

for any $u \geq b$, where κ is given in (5.7) and $\boldsymbol{\rho}$ is defined as

$$\begin{aligned}\boldsymbol{\rho} &= -\kappa\tilde{\boldsymbol{\vartheta}} + \boldsymbol{\nu}_b + (\kappa - \Phi(b))\boldsymbol{\alpha}_{r_1} \\ &\quad + \left(\frac{\kappa(\mathbb{E}[I(e(\theta))] + \mathbb{E}[\mathscr{E}_{r_1}(e(\theta))]) + \mathbb{E}[\mathscr{S}(e(\theta))]}{1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]}\right. \\ &\quad \left. - \mathbb{E}[\Phi(e(\theta + b))]\right)\boldsymbol{\alpha}_{r_2} + \boldsymbol{\alpha}_{\Phi}.\end{aligned}$$

Here, $\Phi(x)$ is given in (6.2) and $\mathbb{E}[I(e(\theta))]$, $\mathbb{E}[\mathscr{S}(e(\theta))]$, $\mathbb{E}[\mathscr{E}_{r_k}(e(\theta))]$, $k = 1, 2$ in (6.3).

Moreover, there exist constants $\{C_j(\theta, b);\ j = 1, 2\}$ such that

$$\begin{aligned}\Psi(u; b) &= C_1(\theta, b)e^{-\xi_1(b-u)} + C_2(\theta, b)e^{-\xi_2(b-u)} + \boldsymbol{\pi} e^{\mathbf{T}u}\mathbf{1} \\ &\quad + \frac{1}{q}(\boldsymbol{\nu}_\omega - \omega(0)\boldsymbol{\vartheta})e^{\mathbf{V}u}\mathbf{1}\end{aligned} \tag{6.6}$$

for $0 \leq u \leq b$, where $\boldsymbol{\pi} \triangleq \omega_1/q\boldsymbol{\alpha} + \sum_{j=1}^{2} C_j(\theta, b)e^{-\xi_j b}\boldsymbol{\alpha}(\xi_j\mathbf{I} - \mathbf{T})^{-1}\mathbf{T}/(\boldsymbol{\alpha}(\xi_j\mathbf{I} - \mathbf{T})^{-1}\boldsymbol{t})$.

Proof. When $P \sim PH(\boldsymbol{\mu}, \mathbf{A})$, we can simplify $w_1(u)$ and $\mathscr{S}(u)$ as

$$\begin{aligned}w_1(u) &= \frac{1}{\tilde{q}}\left\{(\tilde{q} - \boldsymbol{\rho}_1\mathbf{1})\boldsymbol{\alpha} e^{\tilde{\mathbf{T}}u}\mathbf{1} + \boldsymbol{\rho}_1 e^{\tilde{\mathbf{V}}u}\mathbf{1}\right\} \quad \text{and} \\ \mathscr{S}(u) &= \frac{1}{\tilde{q}}\left\{-\boldsymbol{\rho}_2\mathbf{1}\boldsymbol{\alpha} e^{\tilde{\mathbf{T}}u}\mathbf{1} + \boldsymbol{\rho}_2 e^{\tilde{\mathbf{V}}u}\mathbf{1}\right\}\end{aligned} \tag{6.7}$$

according to Proposition 6.2, where $\boldsymbol{\rho}_2$ is defined in (6.4) and

$$\boldsymbol{\rho}_1 = -\tilde{\boldsymbol{\vartheta}} + \boldsymbol{\alpha}_{r_1} + \frac{\mathbb{E}[I(e(\theta))] + \mathbb{E}[\mathscr{E}_{r_1}(e(\theta))]}{1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]}\boldsymbol{\alpha}_{r_2}.$$

By plugging (6.7) into (5.6), straightforward algebra leads to (6.5).

For $0 \leq u \leq b$, we begin with expressions for functions $\{\upsilon_j(y);\ j = 1, 2\}$ in (4.14). Since (4.10) and (6.1) imply that

$$\mathscr{L}_\xi(y) = \frac{e^{\xi y}\int_y^{\infty} e^{-\xi z} f_{-\underline{X}_{e(\delta)}}(z)\mathrm{d}z}{\phi^-(\xi)} = \boldsymbol{\beta}_\xi e^{\mathbf{T}y}\mathbf{1},$$

where $\boldsymbol{\beta}_\xi \triangleq -\boldsymbol{\alpha}(\xi\mathbf{I} - \mathbf{T})^{-1}\mathbf{T}/\phi^-(\xi) = -\boldsymbol{\alpha}(\xi\mathbf{I} - \mathbf{T})^{-1}\mathbf{T}/\boldsymbol{\alpha}(\xi\mathbf{I} - \mathbf{T})^{-1}\mathbf{t}$, $\{\upsilon_j(y);\ j = 1, 2\}$ can be rewritten as

$$\upsilon_j(y) = e^{-\xi_j(b-y)} - e^{-\xi_j b}\boldsymbol{\beta}_{\xi_j} e^{\mathbf{T}y}\mathbf{1}, \quad j = 1, 2.$$

Furthermore, $\{r_k(u);\ k = 1, 2\}$ are linear combinations of functions $\{\upsilon_j(u);\ j = 1, 2\}$, i.e.

$$r_k(u) = C_{k1}\upsilon_1(u) + C_{k2}\upsilon_2(u),$$

where $\{C_{kj};\ k, j = 1, 2\}$ are obtained by matching the coefficients of $r_1(u)$ and $r_2(u)$ in the above and (4.13). Now, denote

$$\begin{aligned}M &\triangleq \frac{\kappa(\mathbb{E}[I(e(\theta))] + \mathbb{E}[\mathscr{E}_{r_1}(e(\theta))]) + \mathbb{E}[\mathscr{S}(e(\theta))]}{1 - \mathbb{E}[\mathscr{E}_{r_2}(e(\theta))]} \\ &\quad - \mathbb{E}[\Phi(e(\theta + b))],\end{aligned}$$

where $\mathbb{E}[I(e(\theta))]$, $\mathbb{E}[\mathscr{S}(e(\theta))]$ and $\{\mathbb{E}[\mathscr{E}_{r_k}(e(\theta))];\ k = 1, 2\}$ are given in (6.3). Then the EDP function $\Psi(u; b)$ for $0 \leq u \leq b$ in (5.6) is simplified to

$$\begin{aligned}\Psi(u; b) &= \kappa w_2(u) + J(u) \\ &= \Phi(u) + (\kappa - \Phi(b))r_1(u) + r_2(u)M \\ &= \Phi(u) + \{(\kappa - \Phi(b))C_{11} + MC_{21}\}\upsilon_1(u) \\ &\quad + \{(\kappa - \Phi(b))C_{12} + MC_{22}\}\upsilon_2(u) \\ &= C_1(\theta, b)e^{-\xi_1(b-u)} + C_2(\theta, b)e^{-\xi_2(b-u)} + \boldsymbol{\pi} e^{\mathbf{T}u}\mathbf{1} \\ &\quad + \frac{1}{q}(\boldsymbol{\nu}_\omega - \omega(0)\boldsymbol{\vartheta})e^{\mathbf{V}u}\mathbf{1},\end{aligned}$$

where $C_j(\theta, b) = (\kappa - \Phi(b))C_{1j} + MC_{2j}$; $j = 1, 2$. The last equality is obtained from (6.2). The proof is therefore complete. □

7. Concluding remarks

In this paper, we generalize the Cramér–Lundberg risk model with diffusion by incorporating exponential upward jumps and arbitrary downward jumps and the threshold dividend policy. The approach we employ also allows for relaxing the positive

loading condition. We have derived and solved a piecewise integral equation satisfied by the Gerber–Shiu EDP function for the retained surplus process. Furthermore, the expression of the EDP function is simplified and expressed in a matrix form when the downward jumps are phase-type distributed. Although in this paper we focus on the exponential upward jumps, our results should be able to generalize to hyper-exponential upward jumps. However, the expression for the EDP function would be complicated.

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Appendix A. The proof of Proposition 4.3

The proof of Proposition 4.3 is lengthy. In the following, we divide the proof into several lemmas and propositions. First, we rewrite the results of Kadankov and Kadankova (2005) as the following lemma.

Lemma A.1. *(Also see Lemma 1 in Kadankova and Veraverbeke (2007)).*

For a general Lévy process $\{X_t; t \geq 0\}$ with $X(0) = 0$, the Laplace transforms of the joint distribution of the exit time and the corresponding overshoot (τ_u^b, Λ) defined in (4.8) and (4.9) satisfy the following equations

$$\begin{aligned}
&\mathbb{E}[\mathrm{e}^{-\delta\tau_u^b}\mathbb{I}(\Lambda \in \mathrm{d}y, \tau_u^b = \tau_{b-u}^+)] \\
&\quad = f_+^\delta(b-u, \mathrm{d}y) + \int_0^\infty f_+^\delta(b-u, \mathrm{d}v) L_+^\delta(v, \mathrm{d}y), \\
&\mathbb{E}[\mathrm{e}^{-\delta\tau_u^b}\mathbb{I}(\Lambda \in \mathrm{d}y, \tau_u^b = \tau_u^-)] \\
&\quad = f_-^\delta(u, \mathrm{d}y) + \int_0^\infty f_-^\delta(u, \mathrm{d}v) L_-^\delta(v, \mathrm{d}y)
\end{aligned} \tag{A.1}$$

for any $\delta > 0$, $u \in [0, b]$ and $y \in \mathbb{R}_+$, where

$$\begin{aligned}
f_+^\delta(b-u, \mathrm{d}y) &= \mathbb{E}[\mathrm{e}^{-\delta\tau_{b-u}^+}\mathbb{I}(H_X(b-u) \in \mathrm{d}y)] - \int_0^\infty \mathbb{E}[\mathrm{e}^{-\delta\tau_u^-} \\
&\quad \times \mathbb{I}(L_X(u) \in \mathrm{d}v)]\mathbb{E}[\mathrm{e}^{-\delta\tau_{v+b}^+}\mathbb{I}(H_X(v+b) \in \mathrm{d}y)], \\
f_-^\delta(u, \mathrm{d}y) &= \mathbb{E}[\mathrm{e}^{-\delta\tau_u^-}\mathbb{I}(L_X(u) \in \mathrm{d}y)] - \int_0^\infty \mathbb{E}[\mathrm{e}^{-\delta\tau_{b-u}^+} \\
&\quad \times \mathbb{I}(H_X(b-u) \in \mathrm{d}v)]\mathbb{E}[\mathrm{e}^{-\delta\tau_{v+b}^-}\mathbb{I}(L_X(v+b) \in \mathrm{d}y)],
\end{aligned}$$

and $L_\pm^\delta(v, \mathrm{d}y) = \sum_{n=1}^\infty L_\pm^{(n)}(v, \mathrm{d}y, \delta)$, $v \geq 0$ are the series of the successive iterations of kernels $L_\pm(v, \mathrm{d}y, \delta)$. These kernels are given by

$$\begin{aligned}
L_+(v, \mathrm{d}y, \delta) &= \int_0^\infty \mathbb{E}[\mathrm{e}^{-\delta\tau_{v+b}^-}\mathbb{I}(L_X(v+b) \in \mathrm{d}l)] \\
&\quad \times \mathbb{E}[\mathrm{e}^{-\delta\tau_{l+b}^+}\mathbb{I}(H_X(l+b) \in \mathrm{d}y)], \\
L_-(v, \mathrm{d}y, \delta) &= \int_0^\infty \mathbb{E}[\mathrm{e}^{-\delta\tau_{v+b}^+}\mathbb{I}(H_X(v+b) \in \mathrm{d}l)] \\
&\quad \times \mathbb{E}[\mathrm{e}^{-\delta\tau_{l+b}^-}\mathbb{I}(L_X(l+b) \in \mathrm{d}y)],
\end{aligned} \tag{A.2}$$

and their successive iterations are defined by

$$\begin{aligned}
&L_\pm^{(1)}(v, \mathrm{d}y, \delta) = L_\pm(v, \mathrm{d}y, \delta), \\
&L_\pm^{(n+1)}(v, \mathrm{d}y, \delta) = \int_0^\infty L_\pm^{(n)}(v, \mathrm{d}l, \delta) L_\pm(l, \mathrm{d}y, \delta), \quad n \geq 1.
\end{aligned} \tag{A.3}$$

The following proposition simplifies the results of the above lemma for $\{X_t; t \geq 0\}$ defined in (3.1).

Proposition A.1. *Let*

$$\begin{aligned}
\bar{f}_1(v) &= \frac{\theta - \xi_1}{\xi_2 - \xi_1}\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(v+b) \\
&\quad + \frac{\xi_2 - \theta}{\xi_2 - \xi_1}\mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(v+b); \qquad \bar{g}_1(y) = \delta_0(y); \\
\bar{f}_2(v) &= \frac{(\theta - \xi_1)(\xi_2 - \theta)}{\theta(\xi_2 - \xi_1)}(\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(v+b) \\
&\quad - \mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(v+b)); \qquad \bar{g}_2(y) = \theta\mathrm{e}^{-\theta y}
\end{aligned} \tag{A.4}$$

for $(v, y) \in \mathbb{R}_+^2$, where $\mathcal{L}_\beta(x)$ is defined in (4.10).

For the Lévy process $\{X_t; t \geq 0\}$ defined in (3.1), we have

$$L_+^\delta(v, \mathrm{d}y) = \left(\sum_{k=1}^2 \sum_{j=1}^2 a_{kj}^* \bar{f}_k(v)\bar{g}_j(y)\right)\mathrm{d}y, \tag{A.5}$$

where the coefficients $\{a_{kj}^*; k, j = 1, 2\}$ are given by

$$\begin{aligned}
a_{11}^* &= \frac{1 - \langle \bar{f}_2, \bar{g}_2\rangle}{\eta}, \qquad a_{12}^* = \frac{\langle \bar{f}_2, \bar{g}_1\rangle}{\eta}, \\
a_{21}^* &= \frac{\langle \bar{f}_1, \bar{g}_2\rangle}{\eta}, \qquad a_{22}^* = \frac{1 - \langle \bar{f}_1, \bar{g}_1\rangle}{\eta}.
\end{aligned} \tag{A.6}$$

Moreover,

$$L_-^\delta(v, \mathrm{d}y) = \left(\sum_{k=1}^2 \sum_{j=1}^2 a_{kj}^* \tilde{f}_k(v)\tilde{g}_j(y)\right)\mathrm{d}y, \tag{A.7}$$

where $\{(\tilde{f}_k, \tilde{g}_k); k = 1, 2\}$ are given in (4.12),

Proof. From the definitions of $\{(\bar{f}_k, \bar{g}_k); k = 1, 2\}$ in (A.4), it is easy to check

$$\langle \bar{f}_k, \bar{g}_j\rangle = \langle \tilde{f}_k, \tilde{g}_j\rangle, \quad k, j = 1, 2, \tag{A.8}$$

and

$$\begin{aligned}
\bar{f}_1(v) + \bar{f}_2(v) &= \frac{\xi_2(\theta - \xi_1)}{\theta(\xi_2 - \xi_1)}\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(v+b) \\
&\quad + \frac{\xi_1(\xi_2 - \theta)}{\theta(\xi_2 - \xi_1)}\mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(v+b) \\
&\leq \frac{\xi_2(\theta - \xi_1)}{\theta(\xi_2 - \xi_1)}\mathrm{e}^{-\xi_1 b} + \frac{\xi_1(\xi_2 - \theta)}{\theta(\xi_2 - \xi_1)}\mathrm{e}^{-\xi_2 b} \\
&= \mathbb{E}\left[\mathrm{e}^{-\delta\tau_b^+}\right].
\end{aligned} \tag{A.9}$$

The last equality is implied by (4.6).

It follows from (4.4) that $L_+(v, \mathrm{d}y, \delta)$ defined in (A.2) can be written as

$$L_+(v, \mathrm{d}y, \delta) = \left(\bar{f}_1(v)\bar{g}_1(y) + \bar{f}_2(v)\bar{g}_2(y)\right)\mathrm{d}y.$$

Thus the successive iteration operation described in (A.3) is equivalent to

$$L_+^{(n)}(v, \mathrm{d}y, \delta) = \left(\sum_{k=1}^2 \sum_{j=1}^2 a_{kj}^{(n)} \bar{f}_k(v)\bar{g}_j(y)\right)\mathrm{d}y, \tag{A.10}$$

where the sequences of coefficients $\{a_{kj}^{(n)}; n \geq 1, k, j = 1, 2\}$ can be computed recursively by

$$a_{kj}^{(n+1)} = \langle \bar{f}_j, \bar{g}_1 \rangle a_{k1}^{(n)} + \langle \bar{f}_j, \bar{g}_2 \rangle a_{k2}^{(n)} \tag{A.11}$$

with $a_{11}^{(1)} = a_{22}^{(1)} = 1$ and $a_{12}^{(1)} = a_{21}^{(1)} = 0$.

Since

$$\langle \bar{f}_k, \bar{g}_1 \rangle = \bar{f}_k(0) \quad \text{and} \quad \langle \bar{f}_k, \bar{g}_2 \rangle = \mathbb{E}[\bar{f}_k(e(\theta))], \quad k = 1, 2,$$

where $e(\theta)$ is exponentially distributed with parameter θ, the recursive scheme (A.11), together with (A.9), implies that

$$\begin{aligned} a_{k1}^{(n+1)} + a_{k2}^{(n+1)} &= \langle \bar{f}_1 + \bar{f}_2, \bar{g}_1 \rangle a_{k1}^{(n)} + \langle \bar{f}_1 + \bar{f}_2, \bar{g}_2 \rangle a_{k2}^{(n)} \\ &\leq \mathbb{E}\left[\mathrm{e}^{-\delta \tau_b^+} \right] \left(a_{k1}^{(n)} + a_{k2}^{(n)} \right), \end{aligned}$$

which implies $\sum_{n=1}^{\infty} \left(a_{k1}^{(n)} + a_{k2}^{(n)} \right) < \infty$ since $\mathbb{E}\left[\mathrm{e}^{-\delta \tau_b^+} \right] < 1$. In turn, we have $a_{kj}^* = \sum_{n=1}^{\infty} a_{kj}^{(n)} < \infty$ for any $k, j = 1, 2$.

Replacing $a_{k(3-j)}^{(n)}$ by $a_{kj}^{(n)}$ and $a_{kj}^{(n-1)}$ in (A.11), we have

$$\begin{aligned} a_{kj}^{(n+1)} &= a_{kj}^{(n)} (\langle \bar{f}_1, \bar{g}_1 \rangle + \langle \bar{f}_2, \bar{g}_2 \rangle) + a_{kj}^{(n-1)} \\ &\quad \times (\langle \bar{f}_1, \bar{g}_2 \rangle \langle \bar{f}_2, \bar{g}_1 \rangle - \langle \bar{f}_1, \bar{g}_1 \rangle \langle \bar{f}_2, \bar{g}_2 \rangle), \quad n \geq 2, \end{aligned}$$

which leads to

$$\begin{aligned} a_{kj}^* - a_{kj}^{(1)} - a_{kj}^{(2)} &= (\langle \bar{f}_1, \bar{g}_1 \rangle + \langle \bar{f}_2, \bar{g}_2 \rangle)(a_{kj}^* - a_{kj}^{(1)}) + a_{kj}^* (\langle \bar{f}_1, \bar{g}_2 \rangle \\ &\quad \times \langle \bar{f}_2, \bar{g}_1 \rangle - \langle \bar{f}_1, \bar{g}_1 \rangle \langle \bar{f}_2, \bar{g}_2 \rangle). \end{aligned} \tag{A.12}$$

Since $a_{11}^{(1)} = a_{22}^{(1)} = 1$, $a_{12}^{(1)} = a_{21}^{(1)} = 0$ and

$$\begin{aligned} a_{11}^{(2)} &= \langle \bar{f}_1, \bar{g}_1 \rangle, \qquad a_{21}^{(2)} = \langle \bar{f}_1, \bar{g}_2 \rangle, \\ a_{12}^{(2)} &= \langle \bar{f}_2, \bar{g}_1 \rangle, \qquad a_{22}^{(2)} = \langle \bar{f}_2, \bar{g}_2 \rangle, \end{aligned}$$

(A.12) leads to (A.6). Since $L_+^{\delta}(v, \mathrm{d}y) = \sum_{n=1}^{\infty} L_+^{(n)}(v, \mathrm{d}y, \delta)$, (A.5) follows directly from (A.10).

Similarly, we have

$$L_-^{(n)}(v, \mathrm{d}y, \delta) = \left(\sum_{k=1}^{2} \sum_{j=1}^{2} a_{kj}^{(n)} \tilde{f}_k(v) \tilde{g}_j(y) \right) \mathrm{d}y.$$

The same argument applies to $L_-^{\delta}(v, \mathrm{d}y) = \sum_{n=1}^{\infty} L_-^{(n)}(v, \mathrm{d}y, \delta)$ and to obtain (A.7). □

Remark A.1. Kadankova and Veraverbeke (2007) study a compound Poisson process with exponential upward jumps, for which there exists only one positive root of the resulting Lundberg fundamental equation. Therefore, their model is tractable and the closed-form expressions of $L_{\pm}^{\delta}(v, \mathrm{d}y)$ can be obtained. When the non-negative drift and diffusion are incorporated, the above proposition shows that the expression of $L_{\pm}^{\delta}(v, \mathrm{d}y)$ is significantly more complicated than that of Kadankova and Veraverbeke (2007). Specifically, it has four terms rather than one. Thus, if the upward jumps are further generalized to be hyper-exponentially distributed with p.d.f.

$$\sum_{j=1}^{n} \alpha_j \theta_j \mathrm{e}^{-\theta_j x}, \quad x > 0,$$

where $\{0 \leq \alpha_j \leq 1; 1 \leq j \leq n\}$ satisfy $\sum_{j=1}^{n} \alpha_j = 1$ and $\{\theta_j > 0; 1 \leq j \leq n\}$ are distinct, we conjecture that $L_{\pm}^{\delta}(v, \mathrm{d}y)$ consists of $(n+1)^2$ different terms.

We are now ready to prove Proposition 4.3.

Proof of Proposition 4.3. It follows from (4.4) that the function $f_+^{\delta}(b-u, \mathrm{d}v)$ in Lemma A.1 can be simplified to

$$\begin{aligned} f_+^{\delta}(b-u, \mathrm{d}v) &= \left(\frac{\theta - \xi_1}{\xi_2 - \xi_1} \mathrm{e}^{-\xi_1 b} [\mathrm{e}^{\xi_1 u} - \mathcal{L}_{\xi_1}(u)] \right. \\ &\quad \left. + \frac{\xi_2 - \theta}{\xi_2 - \xi_1} \mathrm{e}^{-\xi_2 b} [\mathrm{e}^{\xi_2 u} - \mathcal{L}_{\xi_2}(u)] \right) \bar{g}_1(v) \mathrm{d}v \\ &\quad + \frac{(\theta - \xi_1)(\xi_2 - \theta)}{\theta(\xi_2 - \xi_1)} \{ \mathrm{e}^{-\xi_1 b} (\mathrm{e}^{\xi_1 u} - \mathcal{L}_{\xi_1}(u)) \\ &\quad - \mathrm{e}^{-\xi_2 b} (\mathrm{e}^{\xi_2 u} - \mathcal{L}_{\xi_2}(u)) \} \bar{g}_2(v) \mathrm{d}v \\ &= \left(\tilde{r}_1(u) \bar{g}_1(v) + \tilde{r}_2(u) \bar{g}_2(v) \right) \mathrm{d}v, \end{aligned}$$

where

$$\begin{aligned} \tilde{r}_1(u) &\triangleq \frac{\theta - \xi_1}{\xi_2 - \xi_1} \upsilon_1(u) + \frac{\xi_2 - \theta}{\xi_2 - \xi_1} \upsilon_2(u) \quad \text{and} \\ \tilde{r}_2(u) &\triangleq \frac{(\theta - \xi_1)(\xi_2 - \theta)}{\theta(\xi_2 - \xi_1)} \{ \upsilon_1(u) - \upsilon_2(u) \}. \end{aligned}$$

Here, $\{\upsilon_k(u); k = 1, 2\}$ are defined in (4.14). Therefore, $\mathbb{E}[\mathrm{e}^{-\delta \tau_u^b} \mathbb{I}(\Lambda \in \mathrm{d}y, \tau_u^b = \tau_{b-u}^+)]$ in (A.1) can be simplified as

$$\begin{aligned} &\mathbb{E}[\mathrm{e}^{-\delta \tau_u^b} \mathbb{I}(\Lambda \in \mathrm{d}y, \tau_u^b = \tau_{b-u}^+)]/\mathrm{d}y \\ &= (\tilde{r}_1(u) \bar{g}_1(y) + \tilde{r}_2(u) \bar{g}_2(y)) + \int_0^{\infty} (\tilde{r}_1(u) \bar{g}_1(v) + \tilde{r}_2(u) \bar{g}_2(v)) \\ &\quad \times \frac{L_+^{\delta}(v, \mathrm{d}y)}{\mathrm{d}y} \mathrm{d}v \\ &= \bar{g}_1(y) \{ \tilde{r}_1(u)(1 + a_{11}^* \bar{f}_1(0) + a_{21}^* \bar{f}_2(0)) \\ &\quad + \tilde{r}_2(u)(a_{11}^* \mathbb{E}[\bar{f}_1(e(\theta))] + a_{21}^* \mathbb{E}[\bar{f}_2(e(\theta))]) \} \\ &\quad + \bar{g}_2(y) [\tilde{r}_1(u)(a_{12}^* \bar{f}_1(0) + a_{22}^* \bar{f}_2(0)) \\ &\quad + \tilde{r}_2(u) \left(1 + a_{12}^* \mathbb{E}[\bar{f}_1(e(\theta))] + a_{22}^* \mathbb{E}[\bar{f}_2(e(\theta))] \right)] \\ &= \bar{g}_1(y) [\tilde{r}_1(u) a_{11}^* + \tilde{r}_2(u) a_{21}^*] \\ &\quad + \bar{g}_2(y) [\tilde{r}_1(u) a_{12}^* + \tilde{r}_2(u) a_{22}^*], \end{aligned} \tag{A.13}$$

by using (A.6).

Moreover, from (A.4), we have

$$\begin{aligned} \tilde{r}_1(u) a_{11}^* + \tilde{r}_2(u) a_{21}^* &= r_1(u) \quad \text{and} \\ \tilde{r}_1(u) a_{12}^* + \tilde{r}_2(u) a_{22}^* &= r_2(u), \end{aligned} \tag{A.14}$$

where $r_1(u)$ and $r_2(u)$ are given in (4.13). Plugging (A.14) into (A.13) yields (4.16).

Similarly, the function $f_-^{\delta}(u, \mathrm{d}v)$ can be simplified to

$$\begin{aligned} f_-^{\delta}(u, \mathrm{d}v) &= \mathbb{E}[\mathrm{e}^{-\delta \tau_u^-} \mathbb{I}(L_X(u) \in \mathrm{d}v)] \\ &\quad - \left(\mathcal{C}_1(u) \tilde{g}_1(v) + \mathcal{C}_2(u) \tilde{g}_2(v) \right) \mathrm{d}v, \end{aligned}$$

where $\mathcal{C}_1(u) \triangleq \frac{\theta - \xi_1}{\xi_2 - \xi_1} \mathrm{e}^{-\xi_1 (b-u)} + \frac{\xi_2 - \theta}{\xi_2 - \xi_1} \mathrm{e}^{-\xi_2 (b-u)}$ and $\mathcal{C}_2(u) \triangleq \frac{(\theta - \xi_1)(\xi_2 - \theta)}{\theta(\xi_2 - \xi_1)} [\mathrm{e}^{-\xi_1 (b-u)} - \mathrm{e}^{-\xi_2 (b-u)}]$.

Taking a similar argument as (A.13) and using (A.8), we have

$$\begin{aligned} &\mathcal{C}_1(u) \tilde{g}_1(y) + \mathcal{C}_2(u) \tilde{g}_2(y) + \int_0^{\infty} (\mathcal{C}_1(u) \tilde{g}_1(v) \\ &\quad + \mathcal{C}_2(u) \tilde{g}_2(v)) \frac{L_-^{\delta}(v, \mathrm{d}y)}{\mathrm{d}y} \mathrm{d}v \\ &= \tilde{g}_1(y) [\mathcal{C}_1(u) a_{11}^* + \mathcal{C}_2(u) a_{21}^*] \\ &\quad + \tilde{g}_2(y) [\mathcal{C}_1(u) a_{12}^* + \mathcal{C}_2(u) a_{22}^*]. \end{aligned} \tag{A.15}$$

Moreover, (A.7) and (4.12) imply

$$\begin{aligned}
&\int_0^\infty \mathbb{E}[\mathrm{e}^{-\delta\tau_u^-}\mathbb{I}(L_X(u)\in \mathrm{d}v)]L_-^\delta(v,\mathrm{d}y)\\
&=\left(\frac{\theta-\xi_1}{\xi_2-\xi_1}\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(u)+\frac{\xi_2-\theta}{\xi_2-\xi_1}\mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(u)\right)(a_{11}^*\tilde{g}_1(y)\\
&\quad+a_{12}^*\tilde{g}_2(y))\mathrm{d}y+\frac{(\theta-\xi_1)(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}[\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(u)\\
&\quad-\mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(u)](a_{21}^*\tilde{g}_1(y)+a_{22}^*\tilde{g}_2(y))\mathrm{d}y. \qquad (A.16)
\end{aligned}$$

Using (A.15) and (A.16), $\mathbb{E}[\mathrm{e}^{-\delta\tau_u^b}\mathbb{I}(\Lambda\in\mathrm{d}y,\tau_u^b=\tau_u^-)]$ in (A.1) can be simplified as

$$\begin{aligned}
&\mathbb{E}[\mathrm{e}^{-\delta\tau_u^b}\mathbb{I}(\Lambda\in\mathrm{d}y,\tau_u^b=\tau_u^-)]\\
&=f_-^\delta(u,\mathrm{d}y)+\int_0^\infty f_-^\delta(u,\mathrm{d}v)L_-^\delta(v,\mathrm{d}y)\\
&=\left(\frac{\theta-\xi_1}{\xi_2-\xi_1}\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(u)+\frac{\xi_2-\theta}{\xi_2-\xi_1}\mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(u)-\mathcal{C}_1(u)\right)\\
&\quad\times(a_{11}^*\tilde{g}_1(y)+a_{12}^*\tilde{g}_2(y))\mathrm{d}y+\left(\frac{(\theta-\xi_1)(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}[\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(u)\right.\\
&\quad\left.-\mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(u)]-\mathcal{C}_2(u)\right)(a_{21}^*\tilde{g}_1(y)+a_{22}^*\tilde{g}_2(y))\mathrm{d}y\\
&\quad+\mathbb{E}[\mathrm{e}^{-\delta\tau_u^-}\mathbb{I}(L_X(u)\in\mathrm{d}y)]\\
&=\mathbb{E}[\mathrm{e}^{-\delta\tau_u^-}\mathbb{I}(L_X(u)\in\mathrm{d}y)]-\{\tilde{r}_1(u)(a_{11}^*\tilde{g}_1(y)+a_{12}^*\tilde{g}_2(y))\\
&\quad+\tilde{r}_2(u)(a_{21}^*\tilde{g}_1(y)+a_{22}^*\tilde{g}_2(y))\}\mathrm{d}y\\
&=\mathbb{E}[\mathrm{e}^{-\delta\tau_u^-}\mathbb{I}(L_X(u)\in\mathrm{d}y)]-\left(r_1(u)\tilde{g}_1(y)+r_2(u)\tilde{g}_2(y)\right)\mathrm{d}y,
\end{aligned}$$

where the last equality is derived by (A.14). The proof is complete. □

Appendix B. The proof of the formula (4.15)

According to the definitions of $\{\upsilon_j(u);\ j=1,2\}$ in (4.14), it is trivial that

$$\upsilon_j(0)=0,\qquad \upsilon_j(b)=1-\mathrm{e}^{-\xi_j b}\mathcal{L}_{\xi_j}(b). \qquad (B.1)$$

Recall that $\{r_k(u);\ k=1,2\}$ are linear combinations of functions $\{\upsilon_j(u);j=1,2\}$, then we have

$$r_1(0)=r_2(0)=r_2(b)=0,$$

by using (4.13).

We now show that $r_1(b)=1$. Note that

$$\langle\tilde{f}_1,\tilde{g}_1\rangle=\frac{\theta-\xi_1}{\xi_2-\xi_1}\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(b)+\frac{\xi_2-\theta}{\xi_2-\xi_1}\mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(b) \qquad (B.2)$$

and

$$\begin{aligned}
\langle\tilde{f}_2,\tilde{g}_2\rangle&=\frac{(\theta-\xi_1)(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}(\mathrm{e}^{-\xi_1 b}\mathbb{E}[\mathcal{L}_{\xi_1}(b+e(\theta))]\\
&\quad-\mathrm{e}^{-\xi_2 b}\mathbb{E}[\mathcal{L}_{\xi_2}(b+e(\theta))]), \qquad (B.3)
\end{aligned}$$

then we have

$$\begin{aligned}
&\langle\tilde{f}_1,\tilde{g}_1\rangle\langle\tilde{f}_2,\tilde{g}_2\rangle-\langle\tilde{f}_2,\tilde{g}_1\rangle\langle\tilde{f}_1,\tilde{g}_2\rangle\\
&=(\langle\tilde{f}_1,\tilde{g}_1\rangle+\langle\tilde{f}_2,\tilde{g}_1\rangle)\langle\tilde{f}_2,\tilde{g}_2\rangle-\langle\tilde{f}_2,\tilde{g}_1\rangle(\langle\tilde{f}_2,\tilde{g}_2\rangle+\langle\tilde{f}_1,\tilde{g}_2\rangle)\\
&=\langle\tilde{f}_1+\tilde{f}_2,\tilde{g}_1\rangle\langle\tilde{f}_2,\tilde{g}_2\rangle-\langle\tilde{f}_1+\tilde{f}_2,\tilde{g}_2\rangle\langle\tilde{f}_2,\tilde{g}_1\rangle\\
&=\frac{(\theta-\xi_1)(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}\left\{\left(\frac{(\theta-\xi_1)\xi_2}{\theta(\xi_2-\xi_1)}\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(b)\right.\right.\\
&\quad\left.+\frac{(\xi_2-\theta)\xi_1}{\theta(\xi_2-\xi_1)}\mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(b)\right)\\
&\quad\times\left(\sum_{j=1}^2(-1)^{j+1}\mathrm{e}^{-\xi_j b}\mathbb{E}[\mathcal{L}_{\xi_j}(b+e(\theta))]\right)\\
&\quad-\left(\frac{(\theta-\xi_1)\xi_2}{\theta(\xi_2-\xi_1)}\mathrm{e}^{-\xi_1 b}\mathbb{E}[\mathcal{L}_{\xi_1}(b+e(\theta))]+\frac{(\xi_2-\theta)\xi_1}{\theta(\xi_2-\xi_1)}\mathrm{e}^{-\xi_2 b}\right.\\
&\quad\left.\left.\times\mathbb{E}[\mathcal{L}_{\xi_2}(b+e(\theta))]\right)\times(\mathrm{e}^{-\xi_1 b}\mathcal{L}_{\xi_1}(b)-\mathrm{e}^{-\xi_2 b}\mathcal{L}_{\xi_2}(b))\right\}\\
&=\frac{(\theta-\xi_1)(\xi_2-\theta)}{\theta(\xi_2-\xi_1)}\mathrm{e}^{-(\xi_1+\xi_2)b}(\mathcal{L}_{\xi_2}(b)\mathbb{E}[\mathcal{L}_{\xi_1}(b+e(\theta))]\\
&\quad-\mathcal{L}_{\xi_1}(b)\mathbb{E}[\mathcal{L}_{\xi_2}(b+e(\theta))]),
\end{aligned}$$

where the third equality is derived by $\tilde{f}_1(x)+\tilde{f}_2(x)=\frac{(\theta-\xi_1)\xi_2}{\theta(\xi_2-\xi_1)}\mathrm{e}^{-\xi_1(b+x)}+\frac{(\xi_2-\theta)\xi_1}{\theta(\xi_2-\xi_1)}\mathrm{e}^{-\xi_2(b+x)}$.

Plugging (B.1) into (4.13), with the help of (B.2), (B.3) and the above equation, straightforward algebra implies

$$\begin{aligned}
r_1(b)&=\frac{1}{\eta}(1-\langle\tilde{f}_1,\tilde{g}_1\rangle-\langle\tilde{f}_2,\tilde{g}_2\rangle+\langle\tilde{f}_1,\tilde{g}_1\rangle\\
&\quad\times\langle\tilde{f}_2,\tilde{g}_2\rangle-\langle\tilde{f}_2,\tilde{g}_1\rangle\langle\tilde{f}_1,\tilde{g}_2\rangle)=1,
\end{aligned}$$

where the last equality is obtained by the definition of η in (4.11). The proof is therefore complete.

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